ALPACA AI TRADING AGENTS HACKATHON · PAPER TRADING

Theo is a self-improving options-trading agent.

Every cycle it gathers research, forms a falsifiable thesis, simulates the ways to trade it, and sizes the one it trusts most by a track record it has to *earn*. Then it scores itself honestly — whether the **view** was right, whether the **structure** was right, or whether it just got lucky — and only the first two ever move its confidence. That's the self-improving part: not a bigger model, a more honest one.

trdrbot · built on Alpaca's MCP server · $100,000 paper account

THE PROBLEM

Over one week, profit and loss cannot tell skill from luck.

Most trading agents score themselves on P&L. We measured what that actually buys you over a hackathon-length window — and it is close to nothing.

69% of the time a genuinely skilled agent with a **60% edge** out-scores a coin flip over 20 trades. Barely better than a coin flip about the coin flip.	**−7.8% … +8.2%** the range a **zero-skill** agent lands in over the same window. Any result in that band proves nothing at all.	**1 in 3** roughly the chance a coin flip beats real skill. So a leaderboard over one week ranks variance, not ability.

Worse: an agent that *learns* from P&L alone reinforces whatever story happened to correlate with money. That is how a system acquires a superstition.

THE IDEA

Theo asks two separate questions instead of one.

Was the view right? and **was the way I expressed it right?**

	PROFIT	LOSS
HELD	**Reinforce both** the view and the structure earned it	**The view was fine** the strikes, width or stop were wrong
FAILED	**Learn nothing** this was luck — and it blocks promotion	**Correct the view** the structure was faithful

Only one cell is unambiguous — **held and profit**, top left, genuinely earned. The two amber cells are still useful: a loss is informative whichever way it points, so both feed learning. The red cell is the trap: **a profit on a failed thesis**. P&L-based scoring treats that as strong confirmation. Here it teaches nothing, and it never counts toward sizing Theo up.

SELF-IMPROVING, CONCRETELY

Better calibration is not a dashboard metric. It is permission to bet more.

FEEDBACK LOOP

Outcomes score the claim

Every falsifiable claim is pre-registered before it resolves — traded or not. When it resolves, attribution asks which of the four boxes it lands in.

MEMORY

Memory keeps what was useful

Four stores with four different jobs. Memory credits or decays each block that informed a decision by that decision's *verdict* — never by its P&L.

THE LEVER

Size is the reward

Demonstrated calibration raises the Kelly fraction the agent is allowed. Experience → reliability → larger size. No step rewards mere confidence.

And the deterministic half. The model never executes anything. Code wakes the agent, code runs the gauntlets, code computes Kelly, and code fires the exit rules — every 60 seconds, without asking. The LLM's job is narrow: *form a view and propose ways to express it*. Everything downstream of that is arithmetic that cannot be talked round.

THE LOOP

Five stages, looped so the system can learn from itself.

01
Sense
Look at the world. Prices, positions, news, odds, technicals.
every 60s · no LLM

02
Think
Form a falsifiable view; price at least two ways to express it.
~15 min · LLM + arithmetic

03
Act
Place the multi-leg order, then verify it against broker truth.
Alpaca MCP

04
Learn
Guard the position deterministically; then judge it honestly.
every 60s · no LLM

05
Remember
Route what was learned to the store shaped to hold it.
four stores

attribution → calibration → permission to size

STAGE 01 · SENSE

Look at the world — cheap, automatic, every minute.

What it does

Refreshes account and position state from the broker, pulls news and prediction-market odds, and computes technicals, realised volatility, beta and a bootstrap forecast from each name's own return history.

Why it is built this way

- **No LLM here at all.** Observation is not judgement. Sensing is arithmetic, so it runs every tick for almost nothing.
- **A stop checked hourly is worthless.** The cost of not looking scales with what is at risk; the cost of looking scales with LLM spend. So the cheap half runs constantly and the expensive half rarely.
- **A failure renders itself.** An empty news block means the world was quiet; “(unavailable)” means we could not look. Those are different claims.

SUBSYSTEMS

`analytics`	account, positions, quotes, book greeks
`sensors`	news + Polymarket odds into one inbox
`market_stats`	technicals, realised vol, beta, bootstrap
`reconcile`	our record vs the broker's truth

Measured live: SPY and QQQ correlate at 0.92 — so the research universe was rebuilt for *low mutual correlation*, not familiarity.

STAGE 02 · THINK

Three independent ways to have an idea — then one narrow job for the model.

research

Daily, top-down. Technicals + news + prediction-market odds → a regime page and company dossiers → falsifiable opportunities.

discovery

Bottom-up. *The news nominates the companies.* Every nominee must clear a deterministic gauntlet — technicals, forecast, fundamentals, options liquidity — before an LLM writes anything up.

the muse

Creative collision. Random concepts × news × odds, argued into domino chains, every candidate pre-registered and adversarially gated. The top two graduate.

WHAT THE LLM DECIDES

- A **falsifiable thesis** — a claim with a date and a level
- At least **two structurally different** ways to express it
- An honest **probability**, and a vol view if the trade is about vol

WHAT THE CODE DECIDES

- `simulate_experiments` prices every candidate under **one declared measure** — the thesis's own drift and vol, with the market's pricing beside it
- `size_position` computes Kelly from the **conditional** payoff and shrinks the claim by measured calibration
- A no-op is a logged, legitimate answer. Theo declines far more often than it trades.

STAGE 03 · ACT

Place the order, then double-check it against the broker.

What it does

Submits a genuine multi-leg options order through Alpaca's MCP server, stamps it with an idempotent client order id, then reconciles every leg against what the broker actually filled.

Why it is built this way

- **The model authors every tool argument.** So a guard rewrites the order id before it leaves — without it, idempotency is whatever the model invented.
- **Risk is repriced from the fill**, not from the model's word. Every book cap sums `max_loss_usd`; if that number came from a model and never met a fill, every later cap is denominated in fiction.
- **The system refuses a whole-book close** above one open position, and adopts any orphan it finds at the broker into the managed set rather than just logging it.

THE REAL ORDER

class	`mleg` — 2-leg vertical, one ticket
legs	`SPY260903P00766000` buy_to_open `SPY260903P00758000` sell_to_open
type	limit, net debit **$1.70**, day
id	`theo-spy-bps-20260828-766758`

Every position traces back to its reasoning through one `position_id` — journal, wiki, memory, and back to the broker.

STAGE 04 · LEARN

Guard the position without consulting the agent. Then judge it honestly.

THE GUARD — EVERY 60 SECONDS

Exit rules are the agent's own commitments, executed deterministically. One signal registry: every rule reads a signal, compares to a threshold, debounces. Thesis-level stops watch the **underlying**, not the noisy option mark.

An exit rule stated only in prose does not exist. Percentages mean what a trader means — a percent of the *net debit paid*, not of gross premium. On the wrong base, three of the four mark-based rules this book has ever carried could not fire at all.

THE SCORING — AT THE THESIS HORIZON

Attribution waits for the horizon the claim named, then asks the two questions from slide 3. A profit on a wrong view counts as **luck** and teaches nothing.

Calibration then scores every stated probability with a Brier score and its Murphy decomposition — reliability (am I overconfident?) separated from resolution (do my forecasts actually separate winners from losers?).

Attribution is the distinctive signal, and it is deliberately expensive to earn: promotion past the second rung requires that *most resolved theses were actually explicable*. A book of luck is not competence, however good the P&L looks.

STAGE 05 · REMEMBER

Four stores, one for each kind of knowledge.

journal

Append-only events. What happened, when, and which model said so. The record, never rewritten.

`1,890 rows`

wiki

Stable reference that rewrites itself: position pages, company dossiers, the regime page.

`markdown on disk`

elfmem

Evolving memory with decay and reinforcement — and a ratified constitution of epistemic principles.

`credited by verdict, not P&L`

ledger

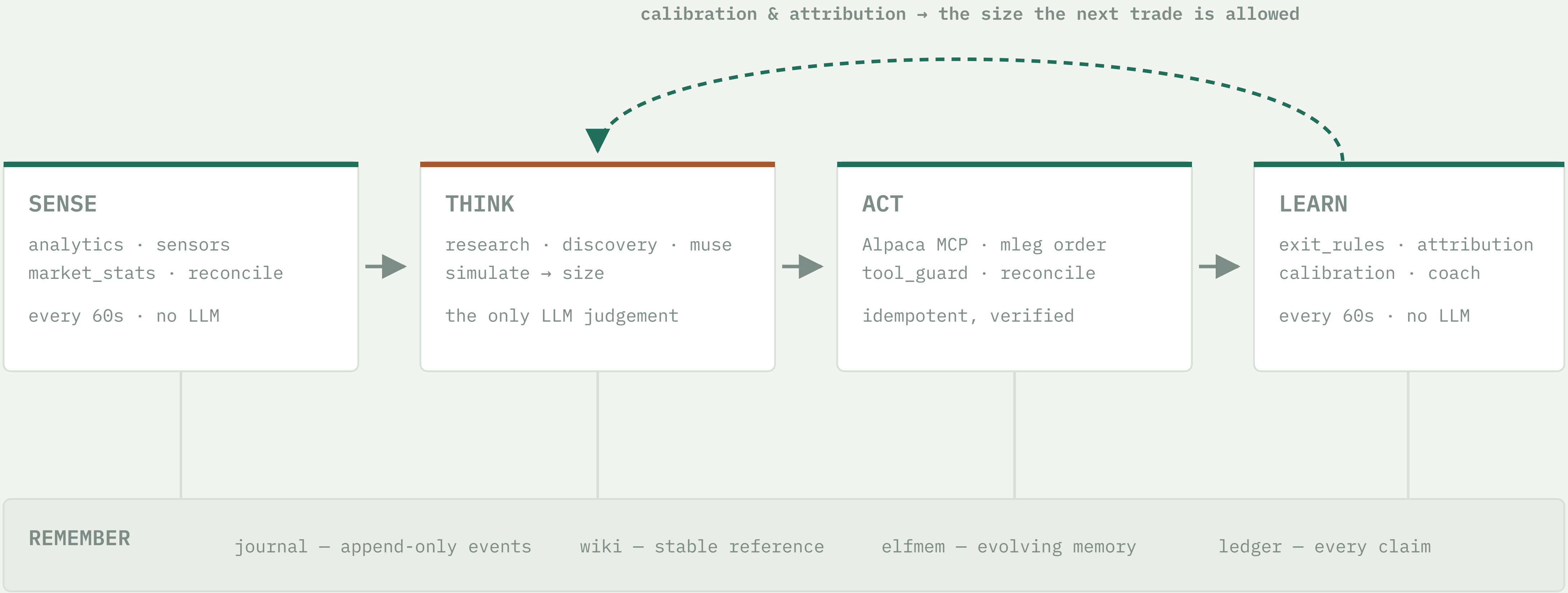
Every falsifiable claim ever made, traded or not — so the trial count a multiple-testing correction needs cannot be lost.

`95 resolved`

The ledger is the quiet one that matters most. **Forecasts on setups we declined are scored too**, at zero capital risk — and they are the only realistic route to a calibration sample that means anything inside a week.

ARCHITECTURE

Where the model sits, and how little of the system it is.



Chassis: one scheduler, a singleton lock, an ordered model fallback chain, and a health probe that asks every subsystem “you ran — but did you *produce* anything?”

Roughly 19,700 lines of Python · 609 offline tests · four trader-readable simulation scaffolds

TECHNOLOGY IMPLEMENTATION

How Alpaca is used — and three things that were harder than they look.

MCP, one session per tick

Alpaca's MCP server runs as a local stdio subprocess. The adapter's default starts a fresh process *per tool call* — six calls cost 12.3s. Sharing one session across the tick cut it to 2.75s.

`−78% wall clock, measured`

Real multi-leg options orders

Verticals, condors and butterflies go as `mleg` tickets with per-leg `position_intent`. Calendars and diagonals are **refused**, not approximated — pricing the far leg needs a model we deliberately do not have.

`a confident wrong payoff is worse than a refusal`

Only 19 of 72 tools bound

Binding all 72 MCP tools cost ~21k tokens of schema *per call*, 71% of it for tools never used once — and a bigger menu measurably worsens tool selection.

`$3.46 → $1.32 per decide cycle`

The lesson that generalises. All three of those had already shipped as code that ran and did nothing: the option-chain compactor failed open against an envelope shape it did not recognise, the session was never actually shared, and the prompt cache was never asked for. Each looked healthy in the logs. That is why `trdrbot health` exists and asks a different question from the tests — *“you ran, but did you produce anything?”*

Something happened in the world, and the equity market had not priced it.

Fed funds futures flipped intraday to price a September *hike* as more likely than a hold. Gold, Bitcoin and TLT all sold off. SPY did not — it had printed 775.30 before the remarks and sat at 769.07, down 0.3%, after a +5.7% run.

That asymmetry is the trade. Not a chart shape — a dated, causal claim that one market had repriced and another had not.

THE THESIS, AS RECORDED

“SPY drifts modestly lower into 2026-09-03 as equities digest fed funds futures repricing.”

horizon 2026-09-03 · falsifiable · pre-registered before any order

SENSE → THINK, THE ACTUAL TOOL CALLS

1. `get_stock_snapshot`
2. `get_stock_bars`
3. `get_clock`
4. `get_news`
5. `get_option_chain`
6. `simulate_experiments`
7. `size_position`
8. `place_option_order`
9. `record_position`

WORKED EXAMPLE · 2 OF 3

Three ways to trade it. The easy-looking one failed the math.

CANDIDATE	WIN RATE	PAYS	RISKS	VERDICT
775/780 call credit spread	76–92%	$95	$283	REJECTED looks safe *because* it usually wins
760 butterfly	low	–	$70 debit	REJECTED needs the fall to stop *exactly* at 760; $22 of 4-leg friction
766/758 put debit spread	42%	2.85 : 1	$2,171	TAKEN EV-positive even at the market's own drift

> "Honest confidence **0.42**. This loses more often than it wins. That is fine — it is the same trade logic as a 38% call spread with 3.8:1 payoff. I resisted rounding up to make it feel better; **the number is scored**."
>
> THEO'S OWN DECISION RECORD, 28 AUGUST

THEN SIZE_POSITION, NOT THE AGENT

- Kelly from the **conditional** payoff — E[win|win] / E[loss|loss], not the max/max ratio, which is measurably biased toward buying premium
- The stated 42% shrunk by measured calibration before it sets size
- Capped three ways at once: per-position, per-underlying, whole book
- **13 contracts. $2,171 max loss — 2.14% of equity.** The agent did not choose that number.

WORKED EXAMPLE · 3 OF 3

Five rules took over. Two questions get answered when the trade closes.

THE GUARD, FROM THE MOMENT OF THE FILL

profit target	+140% of debit
stop	−65% of debit
underlying stop	SPY above 776
time stop	expiry day
leg divergence	broker truth vs ours

The agent named the underlying stop as the one it trusts most: *if SPY makes new highs, the premise that equities haven't discounted the hike is simply false, and the position should die regardless of what the mark says.*

None of the five rules fired — the position closed on an external event before expiry, stop or target.

WHERE IT CLOSED

+129.1%

on net debit. But the P&L is **not** the question that moves anything.

WHAT GETS ASKED ON 3 SEPTEMBER — ONE DAY OUT

- Did SPY close below the level the thesis named? → **was the view right?**
- Given that answer, did this structure express it well? → **was the expression right?**

the horizon hasn't passed yet — a 129% gain still scores as unattributed, not as proof, until it does

RISK

Position size is earned, not chosen — and it is the only reward on offer.

EXPLORE	ESTABLISH	SCALE	MATURE
nothing — the starting allocation	≥5 resolved theses	≥15 resolved, ≥60% *attributable*	≥40 resolved, ≥70% attributable, reliability <0.04
10% BOOK	15% BOOK	20% BOOK	25% BOOK
5% ONE POSITION	7.5% ONE POSITION	10% ONE POSITION	12.5% ONE POSITION

Colour deepens with earned trust — MATURE's tint is the same green this deck uses for a genuinely earned outcome, nowhere else. Climbing costs **evidence**, not confidence: each rung names exactly what must be demonstrated, and none of them can be talked round.

Theo sits on SCALE today, on 76 resolved theses — promotion counts resolved forecasts, not just trades, and it is a different question from whether this week's trades have been attributed yet (next two slides).

- **Monotonic in evidence.** More knowledge can never mean less size. Two separate ladder inversions shipped and were caught by an invariant test that sweeps the whole ladder rather than checking examples.
- **Asymmetric.** Promotion needs a sustained record; a 5% drawdown demotes a whole tier immediately, and 10% goes back to the start.

Attribution is a promotion criterion, not a metric. This is the rung that makes “right vs lucky” cost something real.

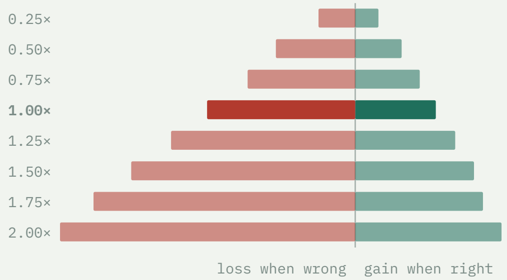
Kelly refuses any structure with unbounded LOSS outright. Unbounded profit is fine — a long call is not a naked short.

RISK / REWARD · DRAG THE SLIDERS

How much risk to take depends on how sure you are the edge is real.

RISK APPETITE 1.00×

BELIEF THE EDGE IS REAL 50%



SPY's own resampled returns, fair-priced structure, the production sizer. Full explorer →

REWARD : RISK

0.54 : 1

median gain when the view is right, against the loss when it is wrong

THE HONEST READING

At **50%** belief the growth-maximising setting is **0.25×**, not 1.00×.

At a coin flip on whether the edge is real, the right setting is the minimum. That is where Theo stands today.

RESULTS · PAPER ACCOUNT · AS OF 2 SEP 2026, 17:26 UTC

The P&L, and why we will not use it as proof.

EQUITY	RETURN	POSITIONS	DECISIONS
$111,415	+11.4%	7	255
from a $100,000 start	high water $111,945 · drawdown 0.5% off it	6 closed, from −2.9% to +129.1% on debit; 1 open, currently −12.6%	157 theses formed, 98 declined outright

Seven positions in a week is still not evidence of skill, and the sharper reason is not sample size. It is that not one of them has been credited as a genuinely explained win yet — the queue that scores view-right-vs-lucky was found stuck this same week (next slide). We report the number because it is real, and we built the machinery precisely so there is something better than this number to judge us on.

THE UNCOMFORTABLE SLIDE

The record says something we would rather it did not. Here it is anyway.

MEASURE	VALUE	READING
resolved forecasts	76	9.8 *effective* — concentrated in a few names
Brier score	0.237	0.25 is a coin flip
reliability	n/a	NOT YET COMPUTABLE too few effective buckets
resolution	0.000	NO DISCRIMINATION forecasts are not separating winners from losers

Resolution is still zero — a stated 70% is not yet landing more often than a stated 55%. Reliability, whether a 40% call actually lands 40% of the time, cannot be computed at all right now: 76 nominal forecasts are only 9.8 *effective* once repeated names are discounted, too thin to bucket by confidence. Two different failures — only a decomposed score tells them apart.

Attribution went the same way, for a sharper reason. The queue that scores a closed position view-right, view-wrong, or lucky was silently stuck for days — found and drained this same week (D-114). Its first-ever verdict: **UNSCOREABLE**, for the one trade with no thesis recorded at entry. Six more, including the +129% position on the previous slide, are still waiting on their own horizon dates. All four boxes from slide 3 currently read zero. The system that would let us hide this is the same system that found it.

76 resolved forecasts clears every count threshold the ladder sets. Concentration, not count, is what still limits reliability — 9.8 effective is the real number.

HONEST LIMITATIONS

What Theo does not do — written down before a judge finds it.

Calendars are refused

Pricing the far leg needs a model we deliberately do not have. A confident wrong payoff is worse than a refusal.

One vol, not a smile

Per-leg IVs set the greeks and the vega-weighted vol, with the EV span across legs printed beside it — but a smile-consistent terminal distribution is refused for the same reason.

The first position can never be attributed

No thesis was recorded at entry. Fabricating one retroactively would be worse than the gap — its permanent verdict is **UNSCOREABLE**, the book's first attribution row.

A decline's structure isn't re-priced

98 declines are each scored as a forecast, at zero capital risk — but the specific spread that got declined is never priced retroactively. "Right about the move" and "right about the trade" stay separate questions for a decline. Logged as an open issue (I-16).

No approval gate, by choice

Paper account, and gates block the feedback loop. What replaces it is stricter: the agent cannot execute a mistake the arithmetic refuses to compute.

Everything above is in the repo

A living bug ledger records every defect the moment it is found and removes it only in the commit that fixes it. 123 numbered issues so far.

IN ONE SENTENCE

Any agent can make money for a week.
Theo can tell you whether it deserved to.

The distinctive thing here is not the strategy. It is that every claim is written down before it resolves, scored afterwards on whether the *reasoning* held, and allowed to change how much capital the next decision gets — with a lucky win worth exactly nothing.

Live record	trdrbot.com — every trade, decline and forecast, updated continuously
Deep dive	the risk research note · the interactive explorer
Repo & demo URL	submitted with this entry's application form
Account	Alpaca paper · $100,000 start